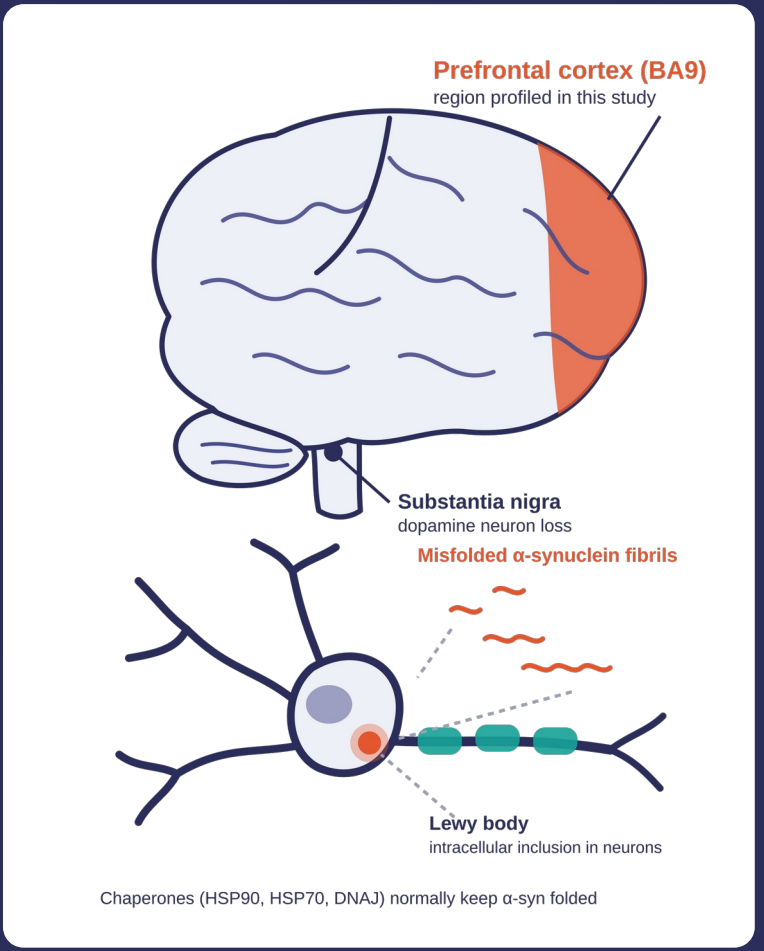
JOURNAL CLUB · PARKINSON'S DISEASE

Single-cell transcriptomic and proteomic analysis of Parkinson's disease brains

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Yale School of Medicine · Science Translational Medicine, 30 Oct 2024

snRNA-seq + proteomics + Lewy pathology of the prefrontal cortex, 6 PD vs 6 controls



Background: PD pathology is known, its cellular architecture is not

A common movement disorder with a classic signature

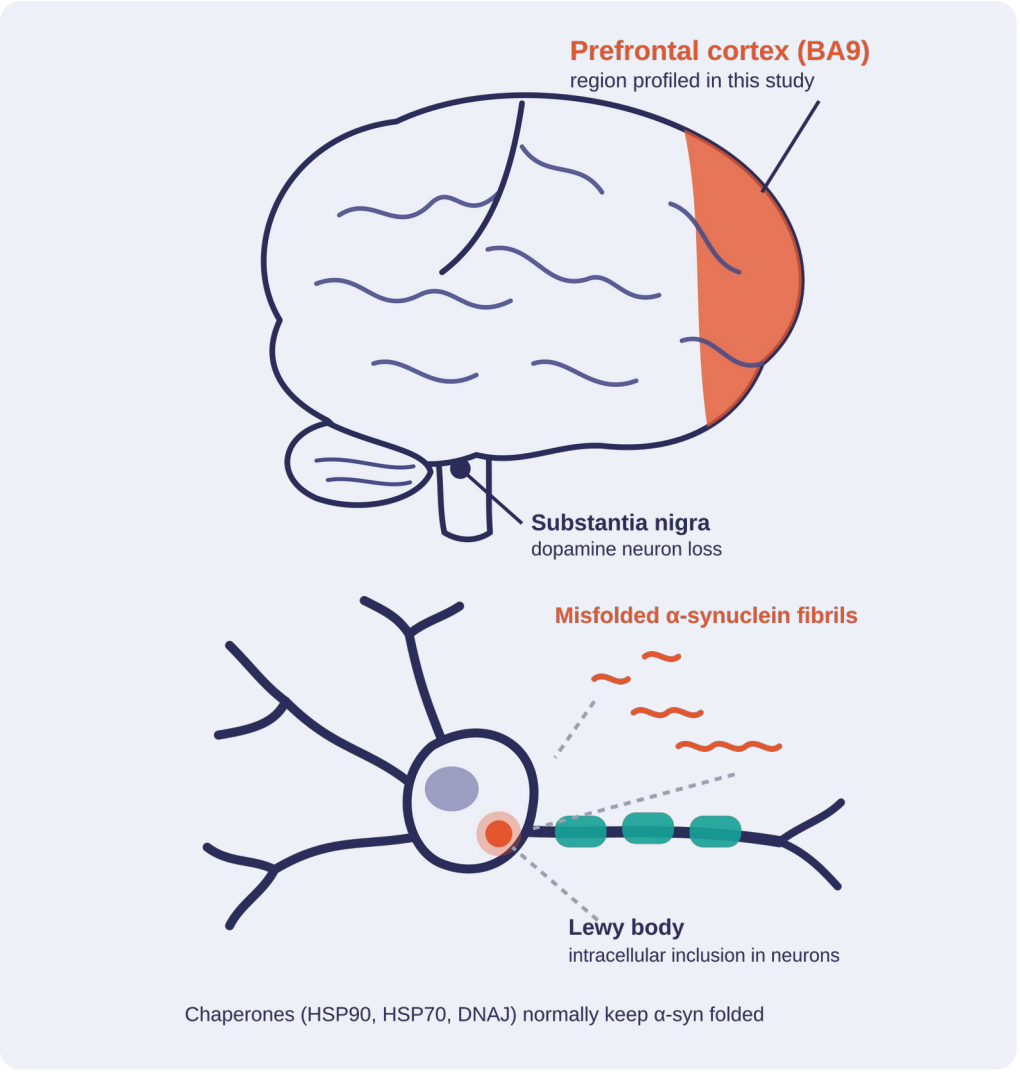
~1 million people in the US and ~6 million worldwide. Hallmarks: loss of substantia nigra dopamine neurons and intracellular α-synuclein aggregates (Lewy bodies).

Neuroinflammation is implicated, but which cells drive it?

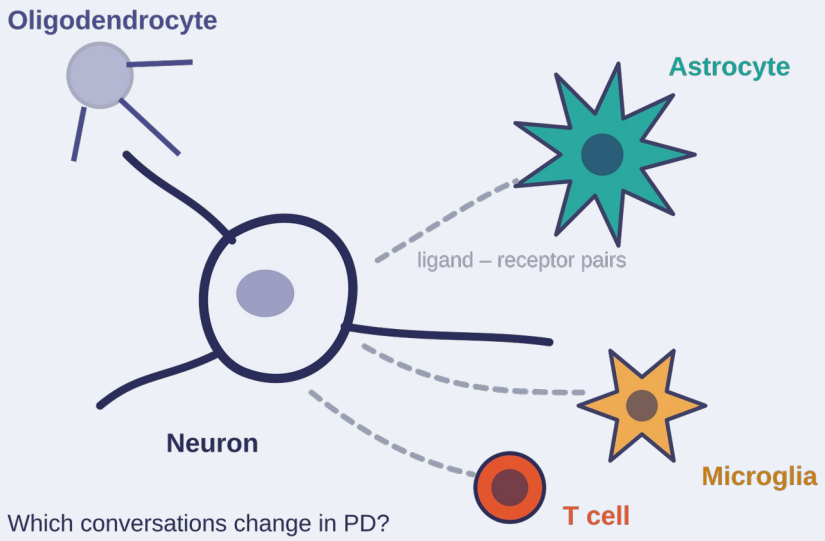
Glial and immune pathways are linked to PD, yet the relative contribution of neurons versus glia to sporadic disease is unresolved. Bulk tissue averages all cell types together.

The cortex was a blind spot

Lewy pathology reaches the prefrontal cortex in late-stage PD, and bulk data flagged this region, but no single-cell map of PD cortex existed. Single-nucleus methods now make postmortem brain tractable.



Objectives: an unbiased, cell-resolved map of PD prefrontal cortex



Central question: which cell types, genes and cell-cell conversations change in late-stage PD cortex, and are they PD-specific?

1 Build a cell-type atlas

snRNA-seq of BA9 from 6 late-stage PD and 6 age-matched controls; identify cell types and their PD-associated DEGs and pathways.

2 Anchor changes to pathology and genetics

Score pS129 α-synuclein Lewy pathology in the same tissue and relate it to neuronal gene expression; place PARK and GWAS genes into cell types.

3 Map crosstalk and the proteome

Infer ligand-receptor communication between neurons, glia and T cells (CellPhoneDB); profile proteins from the same donors and link protein modules to cell types (WGCNA).

4 Ask whether PD looks like AD

Compare DEGs, pathways and proteins against published AD single-nucleus and proteomic datasets from the same cortical region.

Zhu et al., Sci. Transl. Med. 16, eabo1997 (2024)

Methods: three data streams from the same twelve brains

Samples 12 postmortem donors · prefrontal cortex BA9 · 6 late-stage PD vs 6 sex- and age-matched controls (mean 75.9 y)

snRNA-seq (10x Genomics)

Nuclei isolation
Sucrose-gradient ultracentrifugation of frozen tissue; 500-1000 nuclei/µl

→ **Library + sequencing**
Chromium Single Cell 3′ v3; NovaSeq 6000, ~100k reads per nucleus

→ **Alignment**
Cell Ranger 3.1 on a GRCh38 pre-mRNA reference so intronic (unspliced) nuclear reads count

→ **Quality control**
Keep nuclei with 200-7000 genes (drops debris and doublets) → 77,384 nuclei; median 2598 genes

Normalise + feature select
Seurat 4.0.2: log-normalise each sample; top 2000 highly variable genes

→ **Integrate + reduce**
FindIntegrationAnchors → IntegrateData across all 12 donors; PCA → UMAP → 12 clusters

→ **Annotate**
Canonical markers (SLC17A7, GAD1, MBP, AQP4, ITGAM, PDGFRA, CLDN5, CD3E) → 8 cell types

→ **Differential expression**
Wilcoxon rank-sum per cell type (FindMarkers), |log FC| > 0.25, FDR < 0.05; GO via clusterProfiler

Cell proportions
Wilcoxon PD vs HC per type

Variance partition
cell type vs donor, age, sex, PMI

scVelo velocity
spliced vs unspliced, per donor

CellPhoneDB v2
ligand-receptor pairs; 1000 permutations

UTMOST
GWAS → genes → cell-type z-scores

RNAscope
validation in an independent cohort

Proteomics (same tissue)

Label-free LC-MS/MS
35,964 peptides → 2534 proteins across all 12 donors

→ **Differential abundance**
Two-sided t test, P < 0.05 uncorrected → 22 proteins (13 up, 9 down)

→ **WGCNA co-expression**
blockwiseModules, soft threshold 12 ($R^2 \approx 0.9$) → 14 modules, eigengenes PD vs HC

→ **Module → cell type**
One-tailed Fisher exact test vs snRNA-seq marker lists, Bonferroni; GO label per module

Pathology

pS129 α-synuclein IHC
Lewy bodies + neurites counted → pathology score per brain

→ **PD vs HC**
Unpaired t test (P = 0.03); score not linked to RNA quality

→ **Correlate with expression**
Pathology score vs chaperone genes in excitatory neurons (R^2 per gene)

Why these methods?

Pre-mRNA reference: nuclear RNA is largely unspliced; a standard reference would discard most reads.

Anchor integration: removes donor and batch effects while aligning shared cell types, so clusters reflect cell identity (>56% of variance) rather than patient (5.8%).

Wilcoxon rank-sum: non-parametric, no normality assumption for sparse, zero-inflated counts; Seurat default and the same procedure as the AD atlas they compare to.

FDR (Benjamini-Hochberg): thousands of genes tested per cell type; controls the expected false-discovery share.

t test for proteins: log intensities are continuous and roughly normal; with n = 6 vs 6 no protein survives correction, so nominal P is used (a stated limitation).

WGCNA: proteins act in ensembles; unsupervised modules plus Fisher enrichment tie protein changes back to cell types without needing single-cell proteomics.

Permutation test (CellPhoneDB): shuffles cell labels to build an empirical null for each ligand-receptor pair.

Result 1: T cells and microglia expand while neurons go quiet

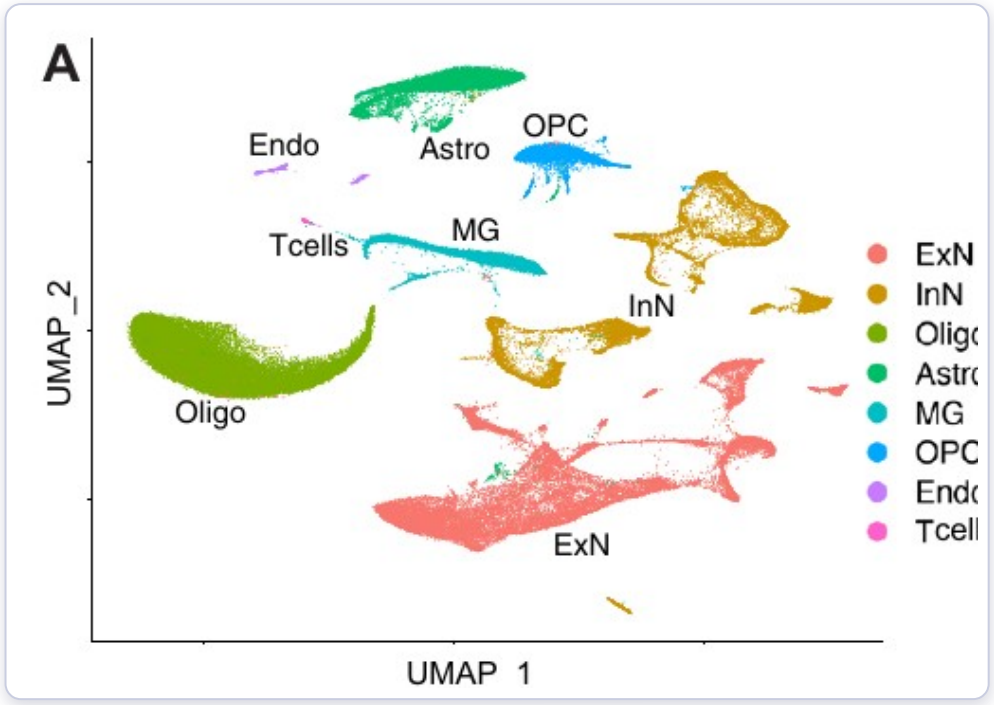


Fig. 1A UMAP of 77,384 nuclei, 12 donors

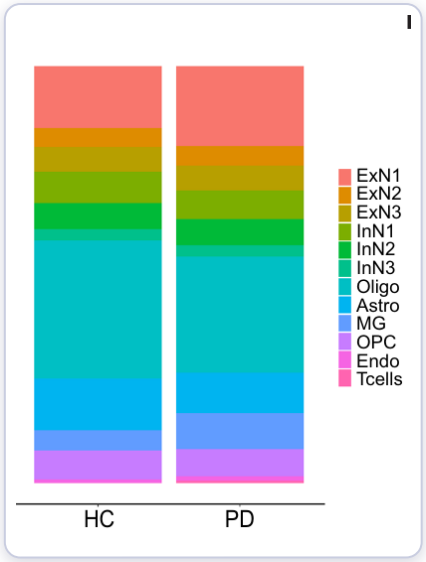


Fig. 1E Cell-type proportions

DEGs	Up	Down
ExN	10	37
InN	4	8
Oligo	48	39
Astro	133	83
MG	178	135
OPC	23	7
Endo	36	22
Tcells	16	27

Fig. 1F DEGs per type

77,384
nuclei profiled (6 PD, 6 HC)

8
major brain cell types

7×
brain-resident T cells in PD (33 → 243 nuclei, P = 0.02)

806
unique DEGs, mostly in astrocytes and microglia

What changed in PD

Proportions: five types stable; T cells up (P = 0.02), microglia up (1699 → 3684, P = 0.09), oligodendrocytes down.

Neurons go quiet: 79% of ExN and 67% of InN DEGs are down, led by protein folding and unfolded-protein response (HSPB1, HSP90AB1, DNAJB6, HSPA1A).

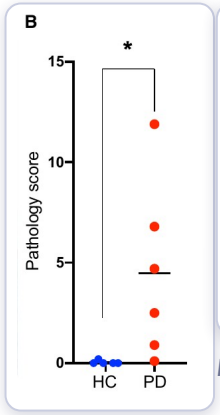
Glia switch on: astrocytes: metal-ion detox (MT1G, MT1F); microglia: myeloid activation and hemopoiesis (FOS, STAT3, RUNX1, IL4R); oligodendrocytes: protein-complex assembly (HSPA1A/B).

T cells: IL7R, IL15, CD28, ZAP70 pathways point to an active adaptive response.

Variance partition: cell type explains >56% of expression variance, donor 5.8%, PMI, age, sex <0.5%.

Result 2: chaperone loss tracks Lewy burden, glial crosstalk fades

Pathology correlation



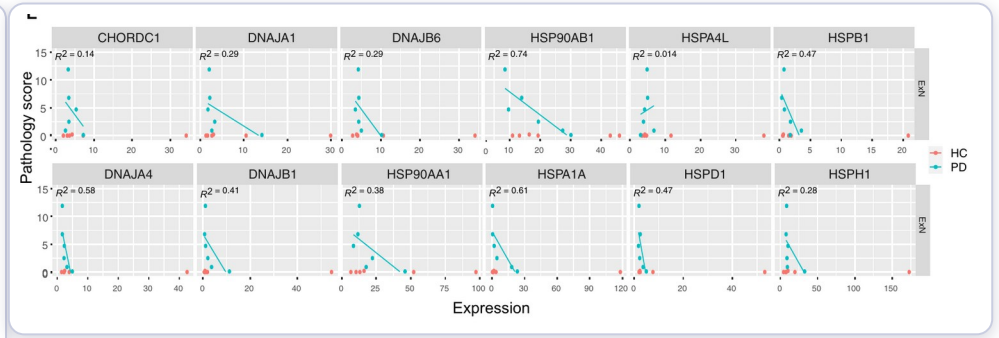


Fig. 2E Pathology score vs chaperone expression in excitatory neurons (teal = PD, red = HC)

Fig. 2B Lewy score

Lewy bodies and neurites are present only in PD cortex (score 0.1-11.9, P = 0.03), with wide donor-to-donor spread.

Higher pathology, fewer chaperones in excitatory neurons: HSP90AB1 R^2 = 0.74, HSPA1A 0.61, DNAJA4 0.58, HSPB1 0.47. HSPA4L (disaggregase) is the one exception.

Inhibitory neurons are spared, consistent with Lewy bodies forming mainly in excitatory neurons.

PD genes: SNCA and LRRK2 up (LRRK2 in microglia and OPCs); PRKN, PINK1, GBA, PARK7 down in excitatory neurons; LRRK2 and GIGYF2 expression scales with pathology.

Cell-cell communication

	Astro	Endo	ExN	InN	MG	Oligo	OPC	Tcells
Astro	−5	8	−38	−15	−4	−2	−20	−12
Endo	8	5	−9	9	5	3	−16	−16
ExN	−38	−9	−17	−16	−11	−13	−30	−29
InN	−15	9	−16	0	−7	−6	−13	−22
MG	−4	5	−11	−7	3	−1	−4	2
Oligo	−2	3	−13	−6	−1	3	−10	−8
OPC	−20	−16	−30	−13	−4	−10	−8	−20
Tcells	−12	−16	−29	−22	2	−8	−20	−3

Fig. 4A Interaction pairs, PD minus HC

Global loss of signalling: nearly every cell-type pair has fewer ligand-receptor interactions in PD; the matrix is almost entirely blue.

ExN-astrocyte hit hardest: ~25% of pairs lost, including EGFR with TGF-α/betacellulin, SIRPA-CD47 (a "don't eat me" signal), TYRO3-GAS6 and activin receptors.

T cells re-wire rather than disconnect: CD74-APP/COPA and CX3CR1-CX3CL1 (immune restraint) are lost; NOTCH1/2-JAG1/2 and TNFRSF1A are gained.

Interpretation: trophic neuron-glia support fades while inflammatory and Notch signalling to T cells emerges.

Result 3: synaptic proteins rise; PD and AD overlap only in glia

Proteomics + WGCNA

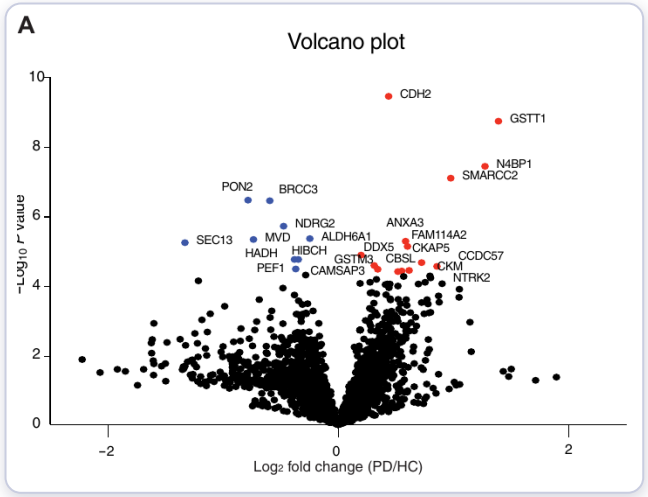


Fig. 5A 2534 proteins, t test

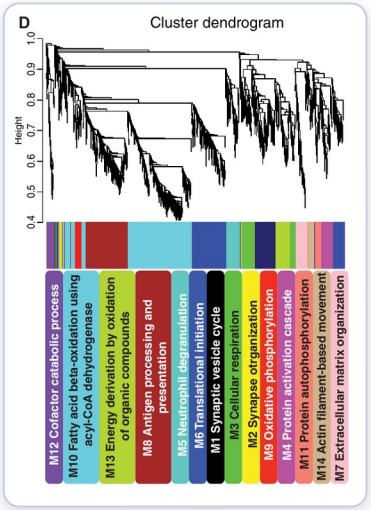
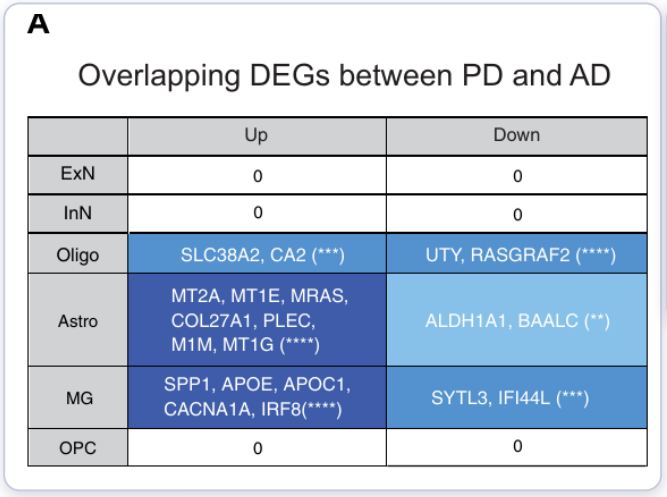


Fig. 5D 14 WGCNA protein modules

PD versus AD



	Up	Down
ExN	0	0
InN	0	0
Oligo	SLC38A2, CA2 (***)	UTY, RASGRAF2 (****)
Astro	MT2A, MT1E, MRAS, COL27A1, PLEC, M1M, MT1G (****)	ALDH1A1, BAALC (**)
MG	SPP1, APOE, APOC1, CACNA1A, IRF8(****)	SYTL3, IFI44L (***)
OPC	0	0

Fig. 6A Shared DEGs with AD (Mathys 2019)

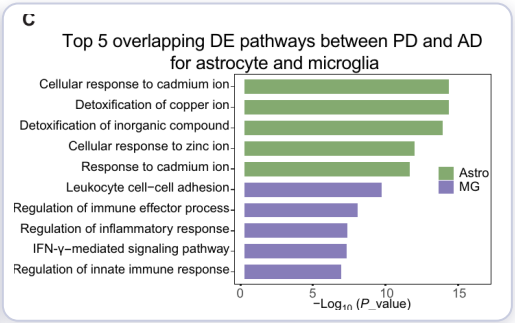


Fig. 6C Shared pathways

22 differential proteins (nominal P < 0.05): CDH2, GSTT1, N4BP1 up; SEC13, BRCC3, NDRG2 down. GO: glutathione metabolism up; branched-chain amino acid catabolism and ER-to-Golgi transport down.

Synapse modules rise in PD: M1 synaptic vesicle cycle and M2 synapse organization have higher eigengene expression and are enriched for excitatory-neuron markers (HPCA, NPTX1, SYN2).

Cross-omics concordance: the same presynaptic assembly and axonogenesis terms were up at the RNA level in neurons, and RNA velocity flagged SYN2.

Zero shared neuronal DEGs between PD and AD prefrontal cortex (24 AD vs 24 controls), despite 138 shared DEGs overall.

Overlap lives in glia: astrocytes share metal-ion detoxification (MT1E, MT2A); microglia share immune activation and APOE, SPP1, APOC1 changes. Oligodendrocytes share SLC38A2, CA2.

Proteome agrees: only CDH2 overlaps with the AD proteome; 12 up and 17 down pathways are shared (cytoskeletal transport, ER-Golgi, actin depolymerization).

Message: neuronal vulnerability is disease-specific; glial inflammation is a common denominator.

Take-aways and critical appraisal

What the paper shows

First single-cell atlas of PD prefrontal cortex (77,384 nuclei, 8 types) with paired proteomics and Lewy pathology from the same donors; data public (GEO GSE202210, PRIDE PXD030142).

Neuroinflammation is cellular and specific: more brain-resident T cells and microglia, activation programs in glia, LRRK2 in microglia/OPCs.

Excitatory neurons are the vulnerable population: chaperone loss scales with α-synuclein burden; HSP90AB1 and DNAJB6, which normally restrain α-syn aggregation, are candidate targets.

Neuron-astrocyte support is lost while neuron-T cell Notch signalling appears.

PD and AD diverge in neurons, converge in glia, arguing for disease-specific neuronal mechanisms on a shared inflammatory background.

Questions to raise in discussion

n = 6 vs 6. Correlations with pathology rest on six PD points; one donor (score 11.9) can drive R^2. Proteomic hits are uncorrected $P < 0.05$ and need replication.

Pseudoreplication. Wilcoxon treats each nucleus as independent, so DEG counts (and the 806 figure) are inflated relative to donor-level pseudobulk tests. Variance partition helps but does not fix this.

Correlation, not causation. Is chaperone loss a cause of aggregation or a late consequence of dying neurons? Only RNAscope-level, not functional, validation.

Design limits. Late-stage, unblinded, end-stage tissue; T cells are rare (276 nuclei); WGCNA pooled PD and HC; AD comparison spans different cohorts, regions (BA9 vs BA10) and pipelines.

Next steps. Larger cohorts with pseudobulk models, early-stage and incidental Lewy body cases, spatial transcriptomics, and perturbing HSP90AB1/DNAJB6 in neuron-glia co-cultures.